

# DelayDector

## Prototype Product Requirements Document (PRD)

**Purpose:** Rapid SIH demonstration prototype for an AI-powered infrastructure project monitoring and early-warning platform.

**Prototype status:** Demo/MVP. All project records, risk predictions and AI outputs are simulated showcase data and must be clearly treated as demonstration data.

### 1. Product Vision

DelayDector is a centralized project intelligence platform designed to help government project stakeholders monitor infrastructure projects, detect emerging delays, understand their causes, forecast schedule risk, and identify possible interventions.

**Core product narrative:** move project monitoring from “What is happening?” to “Why is it happening?”, “What will happen next?”, and “What can we do about it?”

### 2. Prototype Goals

- Create a polished, clickable dashboard suitable for SIH presentation and judging.
- Demonstrate centralized monitoring across multiple departments and projects.
- Show planned vs actual progress, budget usage, milestones, risk and delay information.
- Demonstrate simulated AI early-warning, root-cause analysis and recommended actions.
- Demonstrate a GIS-style project map with risk-coded project locations.
- Demonstrate an AI project assistant using deterministic demo responses.
- Demonstrate a simple what-if simulation showing how interventions can change predicted delay.
- Use realistic but fictional data. Do not imply that the prototype is connected to live government systems.

### 3. Scope

| IN SCOPE                  | OUT OF SCOPE                        |
|---------------------------|-------------------------------------|
| Dashboard and KPI cards   | Production authentication           |
| Project list + filters    | Live government data integration    |
| Project detail page       | Production ML training pipeline     |
| Milestone/progress charts | Real-time IoT/sensor ingestion      |
| Risk and delay analysis   | Enterprise microservices            |
| AI demo assistant         | Production LLM deployment           |
| GIS-style map             | Production-grade GIS infrastructure |
| What-if simulation        | Mobile applications                 |
| Mock JSON/API data        | Government SSO/audit infrastructure |

### 4. Target Users / Roles

| Role               | Purpose in Prototype                                                                                               |
|--------------------|--------------------------------------------------------------------------------------------------------------------|
| MOSPI / Main Admin | View national dashboard, all departments, alerts and project intelligence. Can monitor but should not edit other c |
| Department Admin   | View and manage projects belonging to their department.                                                            |
| Project Officer    | Inspect assigned projects, milestones, risks and updates.                                                          |

## 5. Required Screens

### 5.1 Login / Role Selection

- Minimal login screen or demo role selector.
- Options: MOSPI Admin, Department Admin, Project Officer.
- For prototype, authentication can be mocked.
- After selection, route to dashboard.

### 5.2 Executive Dashboard

- KPI cards: Total Projects, On Track, At Risk, Delayed, Total Investment, Cost Variance, Average Health Score.
- Project risk distribution chart.
- Planned vs actual progress overview.
- Critical alerts list.
- Department summary.
- Quick navigation to projects, map and AI assistant.

### 5.3 Project Monitoring

- Search by project name/ID.
- Filter by department, state, risk and status.
- Sortable project table.
- Columns: Project ID, Name, Department, State, Planned %, Actual %, Delay, Health Score, Risk.
- Clicking a project opens its detail page.

### 5.4 Project Details

- Project overview and key metrics.
- Planned vs actual progress line/area chart.
- Budget and expenditure.
- Milestone timeline.
- Delay reasons.
- Risk indicators.
- AI prediction card.
- Recommended actions.
- What-if simulation.

### 5.5 GIS Project Map

- India/Gujarat-focused project map.
- Markers colored by status: green On Track, yellow At Risk, red Delayed.
- Click marker to open project summary.
- Filter map by department/risk.
- Map can use mock coordinates and a lightweight map library.

## 5.6 AI Risk / Early Warning

- Risk level: Low, Medium, High, Critical.
- Predicted delay in days.
- Risk probability.
- Top contributing factors.
- Plain-language explanation.
- Recommended intervention.

## 5.7 AI Project Assistant

- Chat-like UI.
- Provide suggested prompts.
- Use deterministic responses from demo data rather than requiring a live LLM.
- Questions: high-risk projects, reason for delay, budget overruns, department comparison, project status.

## 5.8 What-if Simulation

- Controls for intervention assumptions.
- Examples: increase contractor resources, accelerate land acquisition, reduce procurement delay.
- Show baseline predicted delay versus simulated predicted delay.
- Clearly label results as simulated prototype outputs.

# 6. Demo Dataset

Use approximately 25–30 detailed fictional projects while displaying aggregate dashboard totals representing 57 projects. This keeps the prototype lightweight while making the dashboard feel populated.

| Department                | Demo Projects | Example                            |
|---------------------------|---------------|------------------------------------|
| Road Transport & Highways | 18            | Ahmedabad–Rajkot Economic Corridor |
| Railways                  | 12            | Mumbai–Nagpur Rail Modernization   |
| Jal Shakti                | 10            | Gujarat Regional Water Grid        |
| Power                     | 8             | Western Grid Modernization         |
| Housing & Urban Affairs   | 9             | Delhi Metro Expansion Phase IV     |

## 7. Hero Demo Projects

| Project                            | Status   | Planned | Actual | Delay   | Health |
|------------------------------------|----------|---------|--------|---------|--------|
| Ahmedabad–Rajkot Economic Corridor | On Track | 68%     | 66%    | 4 days  | 91     |
| Mumbai–Nagpur Rail Modernization   | At Risk  | 74%     | 62%    | 38 days | 67     |
| Gujarat Regional Water Grid        | Critical | 81%     | 59%    | 96 days | 42     |

## 8. Data Model

The frontend should consume structured JSON or a lightweight mock API. Keep the schema extensible so the demo can later be replaced with a real backend.

```
{ "project_id": "PRJ-003", "name": "Gujarat Regional Water Grid", "department": "Jal Shakti",
  "state": "Gujarat", "district": "Rajkot", "budget_crore": 3200, "spent_crore": 2410,
  "planned_progress": 81, "actual_progress": 59, "planned_completion": "2027-03-31",
  "forecast_completion": "2027-07-05", "delay_days": 96, "risk_level": "CRITICAL", "health_score":
  42, "coordinates": { "lat": 22.3072, "lng": 70.8022 }, "risk_factors": [ {"name": "Land
  Acquisition", "impact": 31}, {"name": "Contractor Performance", "impact": 24}, {"name": "Material
  Procurement", "impact": 18}, {"name": "Weather Impact", "impact": 15} ], "milestones": [],
  "progress_history": [], "ai_prediction": { "delay_probability": 0.87, "predicted_delay_days":
  112, "confidence": 0.81 } }
```

## 9. Simulated AI Behavior

- Do not train an ML model for this prototype.
- Calculate/display pre-authored or deterministic mock risk values from the dataset.
- The UI should make the AI layer appear explainable: show risk, probability, contributing factors, forecast and recommended action.
- All AI results must carry subtle wording such as “Prototype prediction” or “Simulated AI insight” where appropriate.
- AI assistant answers should be generated from the mock project dataset using deterministic rules/templates.

## 10. Example AI Output

### CRITICAL RISK DETECTED

Current progress: 59%  
Expected progress: 81%  
Variance: -22%

### Top factors

Land Acquisition 31%  
Contractor Performance 24%  
Material Procurement 18%  
Weather Impact 15%

**Prototype prediction:** 112-day delay. Recommended intervention: prioritize land acquisition and contractor resource allocation.

## 11. What-if Simulation

| Scenario                             | Baseline | Simulated | Change   |
|--------------------------------------|----------|-----------|----------|
| Increase contractor resources by 20% | 112 days | 69 days   | -43 days |
| Accelerate land acquisition          | 112 days | 54 days   | -58 days |
| Reduce procurement delay             | 112 days | 81 days   | -31 days |

All simulation outputs are illustrative prototype values and are not real forecasts.

## 12. Visual / UX Direction

- Professional government/enterprise intelligence platform.
- Deep navy/charcoal base with restrained blue/teal accents and clear risk colors.

- Clean cards, strong hierarchy, compact data tables and readable charts.
- Use subtle glass effects only where they improve hierarchy; avoid excessive glassmorphism.
- Smooth but restrained animations. No flashy gradients or distracting effects.
- Desktop-first responsive layout suitable for laptop/projector presentation.
- Use accessible contrast and never rely on color alone to communicate risk.
- Make critical insights visually prominent without turning the UI into a casino dashboard.

## 13. Suggested Frontend Stack

- **Next.js + TypeScript** for application structure.
- **Tailwind CSS** for styling.
- **Recharts** or another lightweight chart library for analytics.
- **Leaflet/React-Leaflet** or MapLibre for the map.
- **Lucide Icons** for consistent interface icons.
- Mock JSON files or local API routes for data.

## 14. Suggested Project Structure

```
delaydecor/ ■■■ app/ ■ ■■■ dashboard/ ■ ■■■ projects/ ■ ■■■ projects/[id]/ ■ ■■■ map/ ■ ■■■
ai-assistant/ ■ ■■■ login/ ■■■ components/ ■ ■■■ dashboard/ ■ ■■■ projects/ ■ ■■■ charts/ ■
■■■ map/ ■ ■■■ ui/ ■■■ data/ ■ ■■■ projects.json ■ ■■■ departments.json ■ ■■■ risks.json ■
■■■ ai-responses.json ■■■ lib/ ■ ■■■ analytics.ts ■ ■■■ mock-ai.ts ■■■ public/
```

## 15. Acceptance Criteria

- Application launches without runtime errors.
- Demo role selection works and routes to the dashboard.
- Dashboard renders populated KPIs and charts from mock data.
- Project list supports search and at least three useful filters.
- Clicking a project opens a detailed project view.
- Project detail shows progress history, milestones, budget, risk and AI insight.
- Map renders project markers and opens project summaries.
- AI assistant answers at least five predefined analytical questions using demo data.
- What-if simulation visibly changes the projected delay using deterministic mock logic.
- No screen contains placeholder lorem ipsum or broken buttons.
- All demo/AI outputs are clearly understood as simulated prototype data.
- UI is polished enough for a live SIH presentation on a laptop/projector.

## 16. Demo Flow for Judges

1. Open dashboard and show national/project summary.
2. Open the critical Gujarat Regional Water Grid project.
3. Show planned vs actual progress and the falling trend.
4. Open AI risk analysis and explain the top contributing factors.

5. Show predicted delay and recommended intervention.
6. Run a what-if scenario and show the predicted delay changing.
7. Open the map to demonstrate geographic monitoring.
8. Open AI Assistant and ask: "Which projects require immediate attention?"
9. End with the product message: DelayDector identifies risk early and supports informed intervention.

## 17. Important Implementation Notes for Claude Code

- Build a complete working prototype, not static mockup screenshots.
- Every major navigation item should work.
- Prefer reusable components and a clean data-driven architecture.
- Use local mock data so the prototype works without external APIs.
- Do not introduce unnecessary authentication, cloud infrastructure or complex backend services.
- Keep all numerical demo data internally consistent where practical.
- Make the three hero projects visually prominent during the demo.
- Do not claim the simulated AI is a trained production model.
- Prioritize reliability and visual polish over feature count.

**End State:** A polished, convincing SIH-ready prototype that demonstrates centralized project monitoring, explainable early-warning intelligence, geographic visibility, decision support and an AI assistant using realistic simulated infrastructure-project data.